

Concept: Linear equations in one unknown

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Linear equations in one unknown: Transfer

AUTHORED transfer

Subject: Mathematics

Topic: Linear equations

Concept: Operations that preserve the solution set

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Solve a one-unknown linear equation over the rationals

Question type: Constructed response

Provenance: Authored practice

Transfer: Model choice

Solve $3(x - 1) = 5$ over the rationals. Give the exact solution and verify it in the original equation.

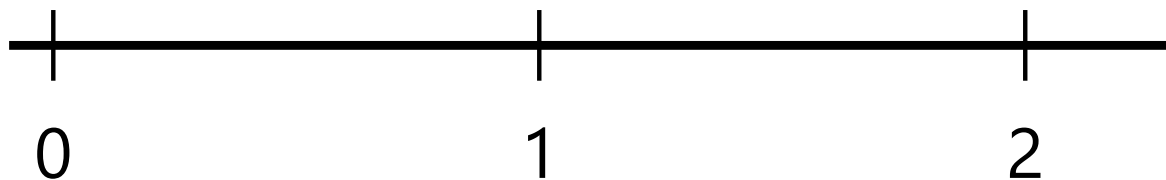
Builds on

- [Solve \$3x + 2 = 9\$ over the rationals. Give the exact solution and verify it.](#)
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Conditions

- The unknown ranges over the rationals.
- Give the solution exactly, not as a rounded decimal.

Domain: the rationals · x (dimension



Declared domain

Where to start

Before touching the equation, say which operation was applied to x last: the subtraction inside the bracket, or the multiplication outside it?

Lineage check

Before you open the solution: what from the earlier item still holds here, and what is different?

